

Messy sources in, a tested KPI page out **every morning**

An ELT pipeline for Copper Kettle, an invented chain of three coffee shops, rebuilt daily by GitHub Actions.

THE PROBLEM

Sales sit in two POS export formats with duplicates and refunds, online orders arrive as webhooks with retries and cancellations, and nobody trusts the weekly spreadsheet built by hand from them.

WHAT I BUILT

- Real daily weather from the Open-Meteo API, with an archive fallback for older days
- Raw files loaded untouched into DuckDB
- dbt staging models into dimensions, facts and a daily store KPI view
- Uniqueness, not-null, relationship, reconciliation and freshness tests
- Daily Actions run that publishes the KPI page, docs and lineage

MEASURED RESULTS

52/52

dbt nodes passed on the first scheduled run, including 40 data tests

4/4

source freshness checks passed

42 s + 10 s

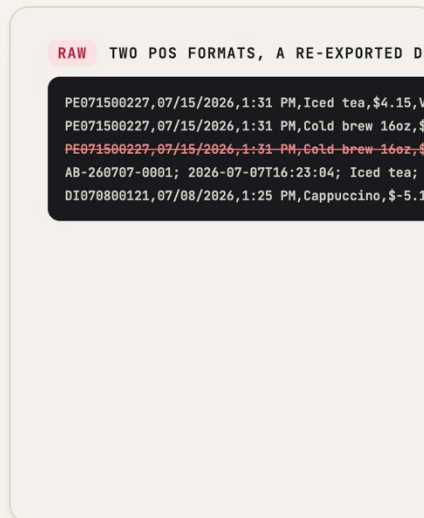
for the pipeline run and the Pages deploy

2 issues

caught by tests while building, both fixed and now tested



The live KPI page



Raw lines to modeled lines



Model lineage

STACK

Python

DuckDB

dbt

GitHub Actions

Open-Meteo API

Live vanthienha199.github.io/data-pipeline-demo/
Code github.com/vanthienha199/data-pipeline-demo